Experiment No: 01
Experiment Name: Introduction to Simulink using MATLAB

1.1 Objectives

- The purpose of this lab report is to familiarize with Simulink and its functionalities.
- It will provide a theoretical overview of Simulink, explaining key concepts and features.
- Additionally, it will present practical examples to illustrate how Simulink can be used to model and simulate different types of systems.

1.2 Introduction

Simulink is a powerful tool in MATLAB that allows engineers and scientists to model, simulate, and analyze dynamic systems. It provides a graphical environment for building and simulating block diagrams, making it an essential tool for various applications such as control system design, signal processing, and communication systems. This lab report aims to provide an introduction to Simulink and its features, along with a demonstration of its capabilities through practical examples.

Simulink allows the construction of block diagrams using a wide range of predefined blocks. These blocks represent mathematical operations, system components, sensors, actuators, and more. By connecting these blocks with signal lines, complex systems can be easily modeled and simulated. Continuous and Discrete Systems: Simulink supports the modeling of both continuous and discrete systems. Continuous systems are represented by differential equations, while discrete systems are represented by difference equations.

1.2.1 Different types of Blocks on Simulink

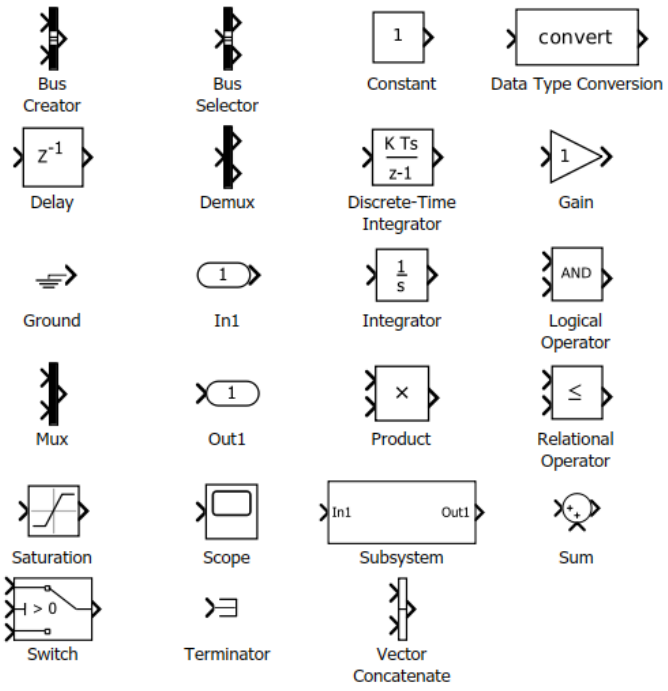


Fig 1.1 : Commonly used Blocks

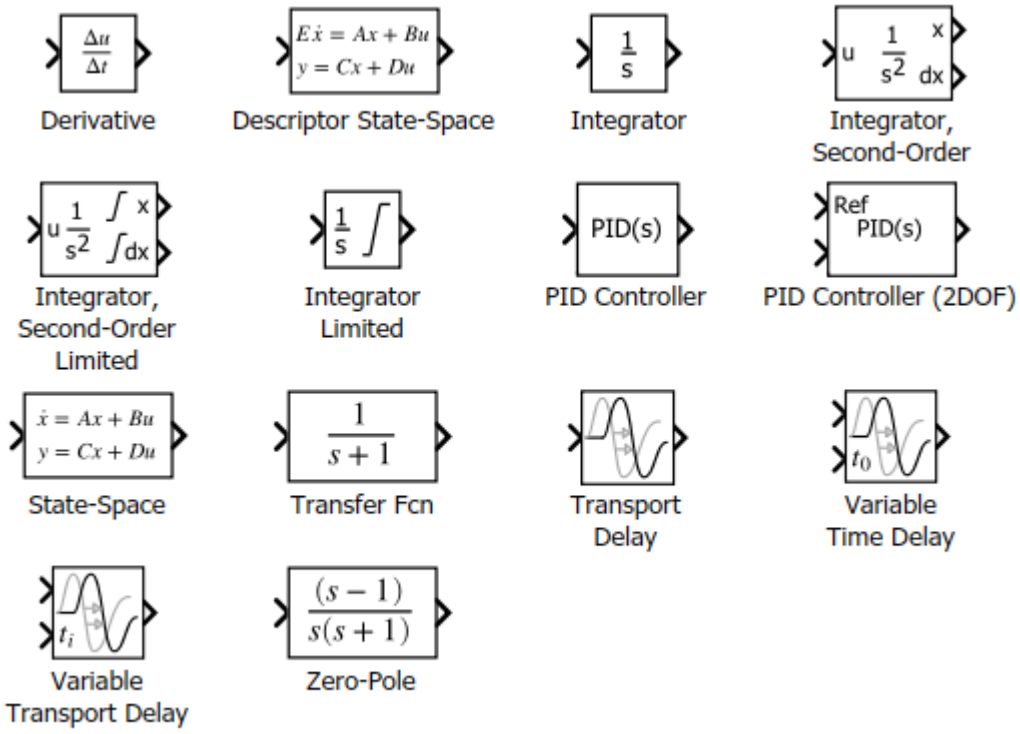


Fig 1.2: Continuous Blocks

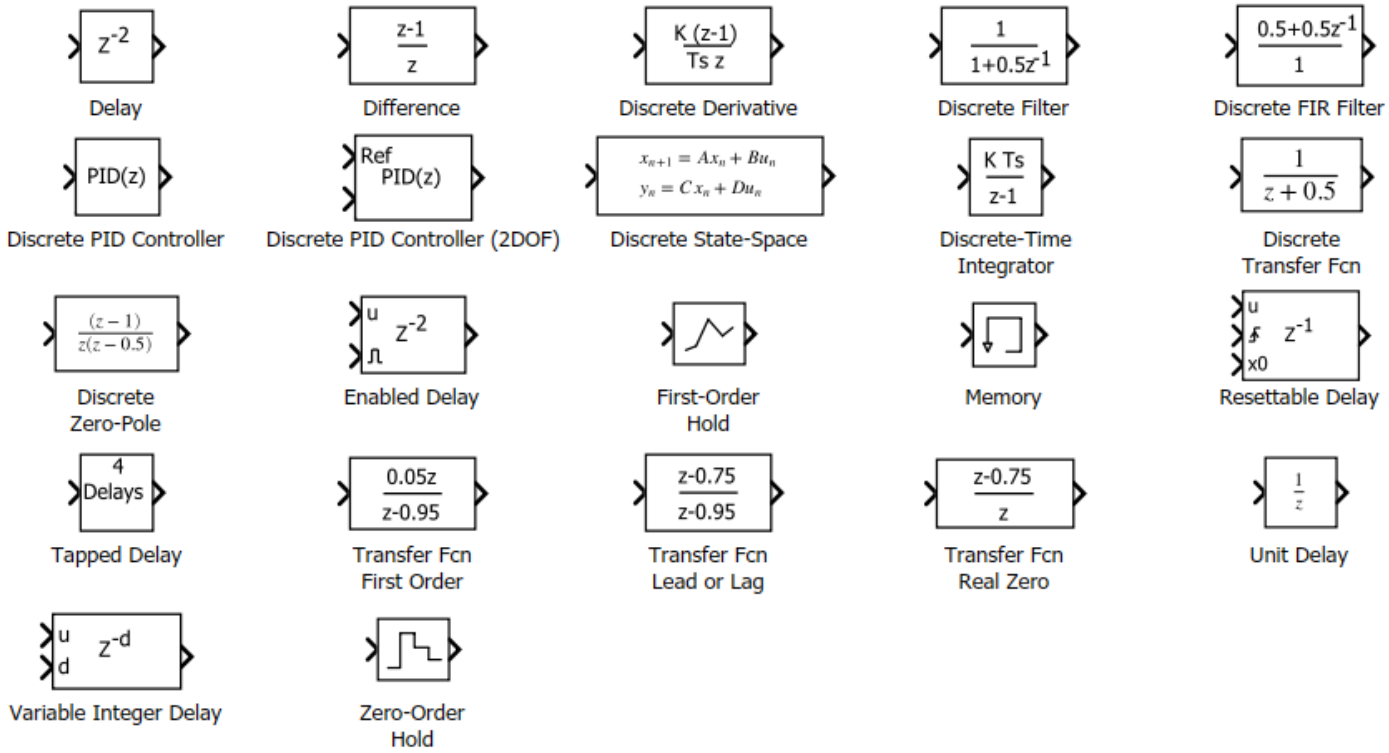


Fig 1.3: Discrete Blocks

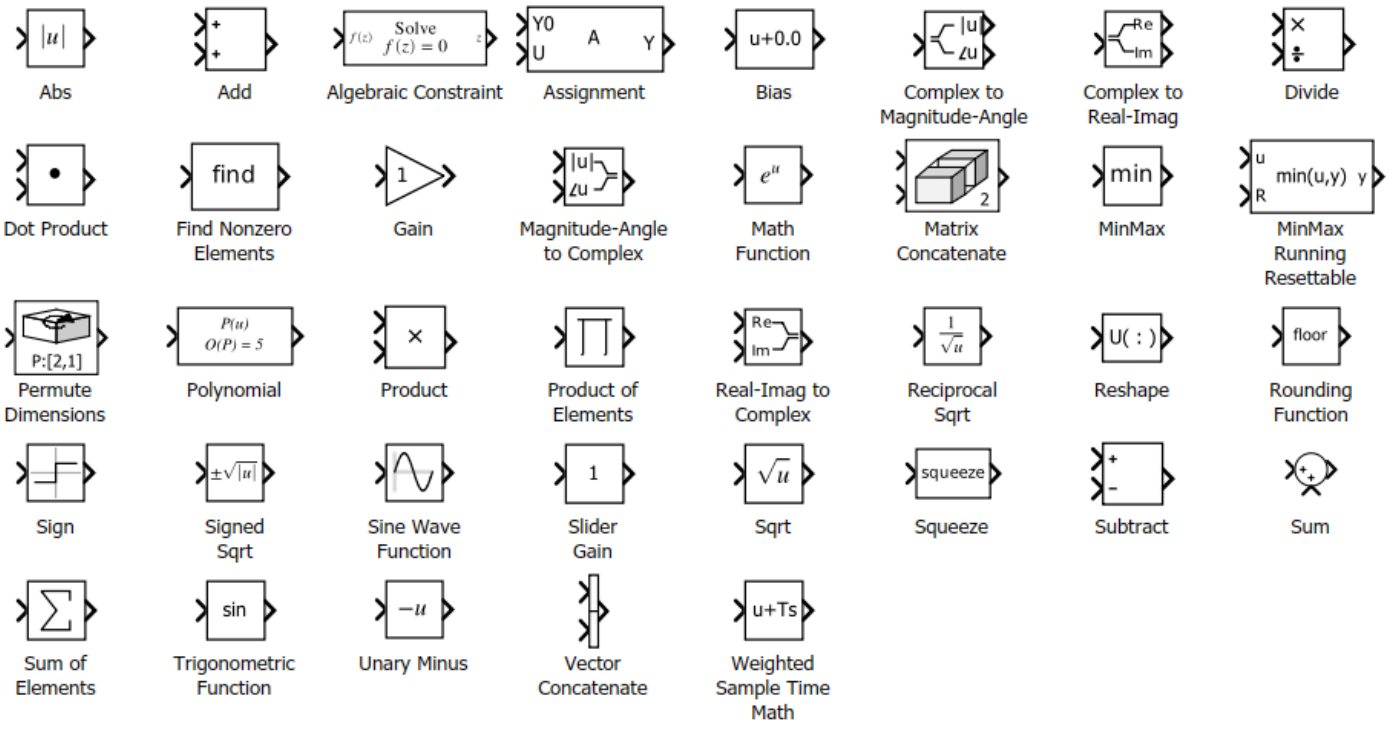


Fig 1.4: Math Operator Blocks

1.3 Circuit Diagram of RLC Circuit

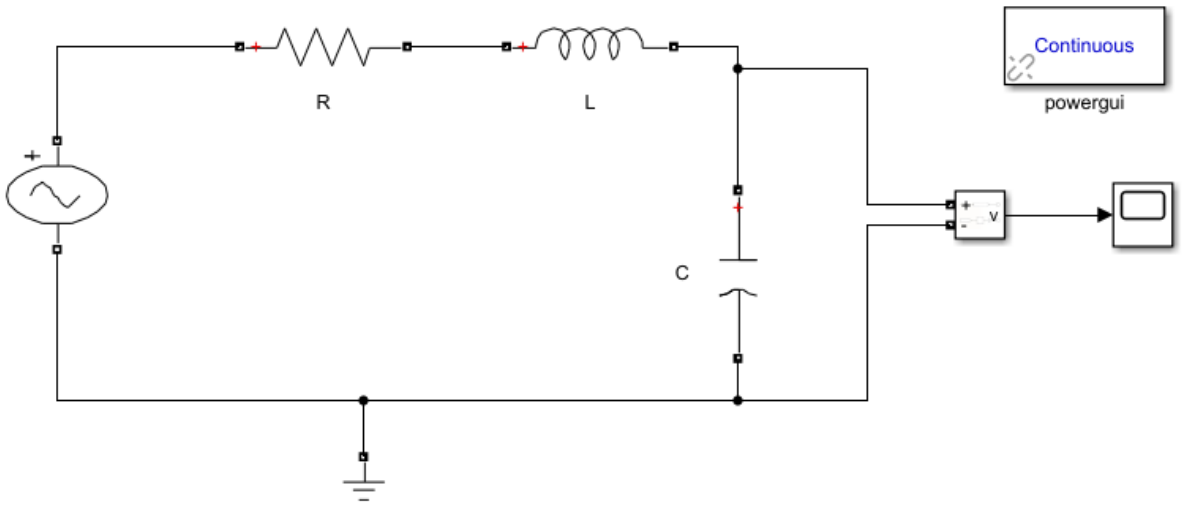


Fig 1.5: Circuit Diagram of RLC Circuit

Equation:

The build up equation for this circuit is

$$\frac{V_o}{V_s} = \frac{1}{1 + S \times R \times C + S^2 \times L \times C} \quad \ldots\ldots\ldots (i)$$

1.4 Circuit Simulation Diagram

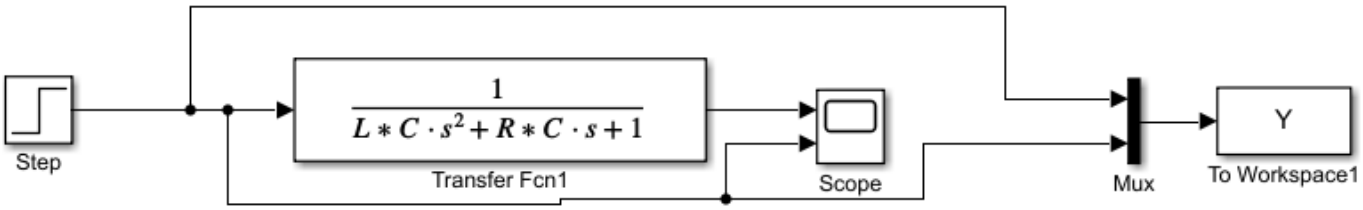


Fig 1.6: Simulation Block in Simulink

1.5 Output Waveform

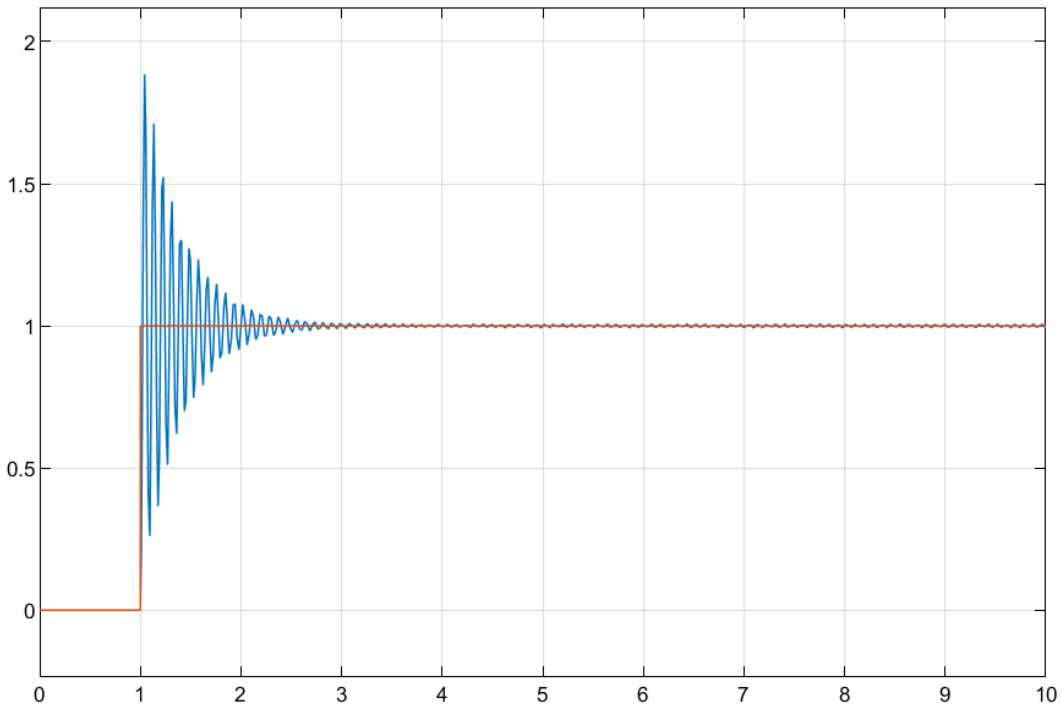


Fig 1.7: Output Response of the System when R = 10Ω, C= 100 uF, L = 2 H

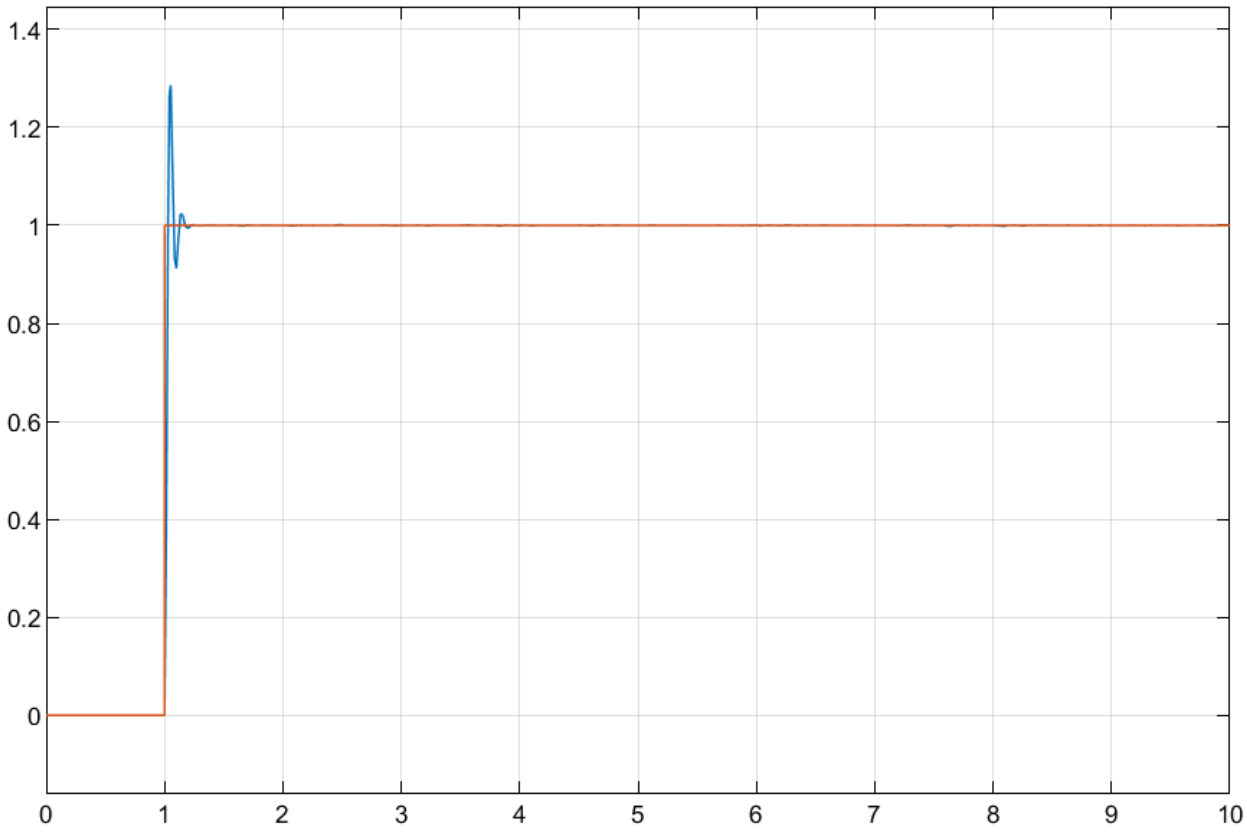


Fig 1.8: Output Response of the System when R = 100Ω, C= 100 uF, L = 2 H

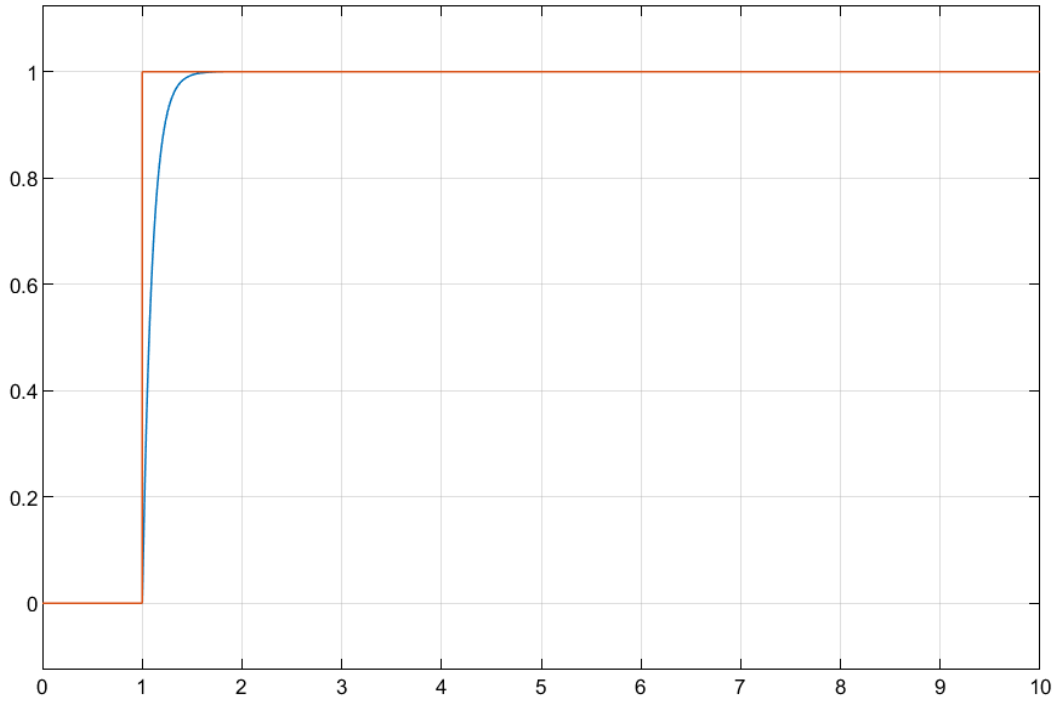


Fig 1.9: Output Response of the System when R = 1 kΩ, C= 100 uF, L = 2 H

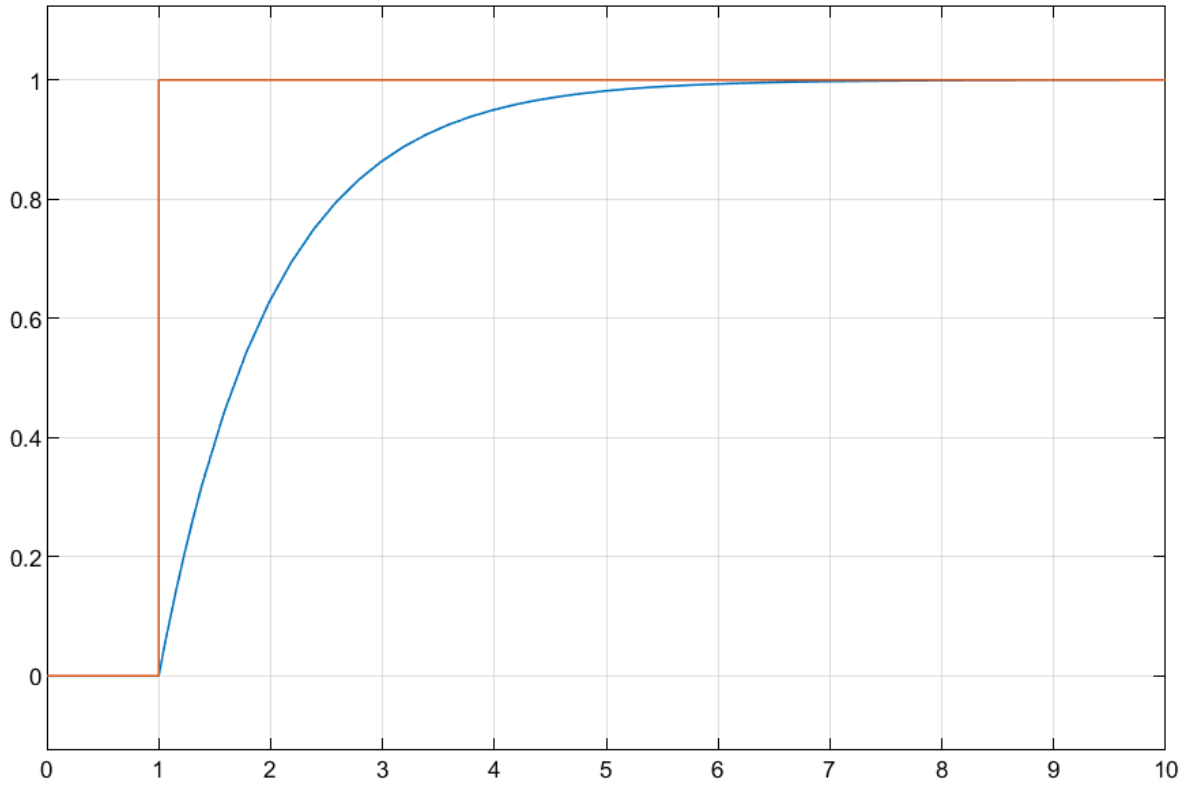


Fig 1.10: Output Response of the System when R = 10 kΩ, C= 100 uF, L = 2 H

1.6 Conclusion

In conclusion, this lab report has provided an introduction to Simulink using MATLAB, demonstrating its capabilities in modeling, simulating, and analyzing dynamic systems. Simulink offers a user-friendly graphical environment for constructing block diagrams, enabling the representation of complex systems. By mastering Simulink, we can enhance ourr system design and analysis capabilities, making it a valuable tool for engineers and scientists. In the RLC Circuit the output is not as same as the input step signal. As increasing the Resistor the output also more smooth and reach to the final position.